ASC

MODEL 980A COMPUTER

MAINTENANCE MANUAL

Volume VIII

LOAD, PIN AND WIRE LISTS

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INTRODUCTION

SECTION I

INTRODUCTION

In order to reduce the number of drawings necessary to properly document the Model 980A electrical configuration; load, pin, and wire lists have been prepared and played-out on a computer print-out. This print-out is designed for use in system evaluation and trouble analysis. This document contains the computer generated load, pin, and backplane wire lists which replace the conventional electrical drawings for the Arithmetic Unit and connector wiring.

This volume does not contain the computer generated lists for the Input/Output (I/O) and I/O Expansion of the Model 980A. The I/O lists are included in Volume IV of the Model 980A Maintenance Manual (Texas Instruments Part No. 960699-9704).

This document contains computer generated load, pin, and backplane wire lists (Sections III, IV, and V, respectively) which replace conventional logic diagrams for the Arithmetic Unit and connector wiring. A description of the load list is included at the beginning of Section III. Similarly, descriptions of the pin and backplane wire lists are included at the beginning of Sections IV and V, respectively.

SECTION II
MAJOR SIGNATURES

SECTION II

MAJOR SIGNATURES

A list of major signatures that are used in the Model 980A Computer System are listed in the following table with the associated functional description.

Signature	Description
ALUB(0-15)	Memory Store Data Lines
ALU(0-15)	Arithmetic Logic Unit Outputs
API(0-15)	Enabled API Data Lines
A(0-15)	Bus A Data Lines
BFR(0-15)	Next ROM Address Bits For An Inst. Fetch
BKPTQ	Breakpoint Condition Indication
BR(0-15)	B Register Contents Are BR0-BR15
BSDQ	Branch, Store, Or Double Length Inst.
BYTFLAG	Denotes Byte Inst. In Process
B(0-15)	Bus B Data Lines
CLKONQ	Synchronized Clock—On Signal
CPUCR	CPU Cycle Request
CPUSTR	CPU Store Request
C(0-15)	Bus C Data Lines
DMAQ	Synchronized DMA Interrupt
DMAREC	Synchronized DMA Interrupt Recognized Signal
DP(0-15)–	I/O Bus Input Data
ER(0-15)	E Register Contents
EXPQ	Synchronized I/O Interrupt
FCLKT–	Free Running AU Clock
FORCE00	Force Next ROM Address To Zero
GCLKA:1	AU Clocks For NAND Gating With TRUE Enables
GCLKB:1	AU Clocks For Direct Use
GCLKC:1–	AU Clocks For Direct Use
GCLKE:1	AU Clocks For Direct Use
GCLKF:1–	AU Clocks For Direct Use
GOIO–	Synchronized I/O GO Strobe
IDLQ	Idle Condition Indication
IFLAGQ	Flag Denoting Interrupt In Process
IOCLK–	External Clock
IOQ	Synchronized I/O Interrupt
IOUT(0-15)	I/O Data Selector
IR(0-15)	Instruction Register Contents
LR(0-15)	L Register Contents

MAJOR SIGNATURES (Continued)

Signature	Description
MAR(0-15)	Memory Address Register Contents
MDR(0-15)	Mem. Data Register Contents
MEM(0-15)	Memory Data Lines
MRESET	Debounced Power Supply Reset Signal
MRESETQ	Synchronized Reset Signal
MR(0-15)	M Register Contents
NRA(0-7)	Next ROM Address
PC(0-15)	Program Counter Contents
PFAIL	Power Failure Is About To Occur
PNLENQ	Synchronized Panel Enable Signal
PNL(0-15)	Console Entry Data
PNLENQ	Synchronized Panel Enable Signal
RAA(0-7)	ROM Address A
RAB(0-7)	ROM Address B
RAC(0-7)	ROM Address C
RDYQ	Synchronized I/O Ready Signal
RESET–	Reset Signal For External Use
RUNFFQ	AU Run Mode Enable Flip-Flop
R(1-72)	Control ROM Outputs
SELBFR	Select BFR As Next ROM Address
SELJ0	Select J0 As Next ROM Address
SELJ1	Select J1 As Next ROM Address
SELJ2	Select J2 As Next ROM Address
SELOP	Select OP Code As Next ROM Address
SELPA	Select PA As Next ROM Address
SR(0-15)	S Register Contents
ST(0-15)	Status Register Contents
TERMIO–	Synchronized I/O Term Strobe
UR(0-15)	Utility Register Contents
XR(0-15)	X Register Contents

SECTION III
LOAD LISTS

SECTION III

LOAD LIST

The function of the load list is to provide quick identification of all pins in any wired chain, including: locations, pin numbers, and semiconductor network types. Each pin of each network is assigned a signal name or pin signature. A load list program has been used to sort all such pin signatures into alpha-numeric order.

All pins with identical pin signatures are grouped together in the list. All such groups of identical pin signatures are wired together to form a chain. Any of all pins of such a chain may be located by referring to the load list.

In the load list all entries are sorted in the following order: (1) pin signature, (2) wiring block number, and (3) location wiring sequence. When sorting by pin signature, the signature characters are read from left to right. Spaces (blanks) are sorted first. Then, letters A through Z have precedence. Special characters, such as the bar symbol (-), have second priority followed by numbers 0 through 9. A sorted list appears below:

IRA 4
IRA4B
IRA4BB
IRA4BB–
IRA4BC–
IRA4B–

The actual load list contains several entries of identical signatures for each example. Further sorting is accomplished within the groups of signatures by block number and location sequence.

To facilitate the design, several diagnostic messages appear throughout the load list. The purpose of the diagnostic message is to indicate errors in original data for the program. All real errors are corrected, and the load lists are updated. Any remaining diagnostic messages are not an indication of an error, but an indication of special cases of normal conditions.

Each group of identical pin signatures is computer analyzed, and the following diagnostic messages may appear:

a. NO SOURCE: This diagnostic message indicates the chain does not contain any pin that has been classified as a source. This message might initially indicate an error as it would seem that each chain should have a source. A common exception is when one pin in the chain is either a connector or a series resistor. Such devices are not classified as a source because the device may also be used as a load.

b. UNUSED: This diagnostic message appears when there is only one entry for a given pin signature and that single entry is a source. It may be assumed that when a source exists, the source should have a load. A common exception is a flip-flop where only one side (normal or inverted) is used.

c. OVERLOADED: Each source drives a specific number of loads. The number of loads is determined by the source and load network types. The driving capability of the source and the current requirements of each load are computed. When the load current exceeds the source capability, this diagnostic message will appear. In all cases, the remaining net load is printed. The net load entry indicates the number of additional loads that can be added before overloading occurs.

d. MULTIPLE SOURCE: Normally, each chain has only one source. Where more than one source appears (sources indicated by asterisk), this diagnostic message will appear. In some circumstances, this will be a normal condition, such as parallel gates. Vcc, ground, etc.

e. EXTERNAL SOURCE: A signal that has only loads and a connector in the chain is designated as an external source.

The load list printout is divided into ten (10) columns. The contents of each column is as follows:

a. Column 1 contains the pin signature. Each pin signature entry represents one pin of a network. All identical pin signatures are sorted together in a group. Each group of identical pin signatures represents a wired chain; i.e. all pins of one group are wired together.

b. Column 2 contains the circuit identification code. This entry is the full network name.

There are also types of devices other than integrated logic networks. These devices are coded: CT for top edge connector pins, CB for bottom edge connector pins (both card edge and rack panel), RES for resistors, and DISC for particular discrete components. DISC is used as a device type where combinations of discrete components are given the same location identification.

c. Column 3 contains the device name. This is an arbitrary name assigned to each network. A device name in conjunction with a pin number (Column 4) uniquely identifies that pin.

d. Column 4 contains the pin name. This is the pin number on the network that locates the pin.

e. Column 5 contains the pin function identification. This column is blank unless the entry is a source in a chain. When the entry is a source, an asterisk appears to identify the entry as a source.

f. Column 6 contains the load identity. The load or driving capability of each pin is shown in tenths of a load. All loads are shown as negative quantities and all sources are shown as positive quantities. Connectors are shown to have zero load. The net (remaining) loads are also shown. If the net load is negative, the diagnostic OVERLOADED identification is normally printed.

g. Column 7 contains the block number. The arithmetic unit, arithmetic option board, and connector plate are separate blocks.

h. Column 8 contains the location name. Each semiconductor device is positioned on a printed circuit board by row and column. Discrete components, such as resistors and capacitors are also positioned by row and column in 14 or 16 pin locations. An example of a location code is 1123. This indicates the package is on the board at row 23 in column 11. The row and column numbers are etched on the component side of the boards.

i. Column 9 is identified as the branch vector or the pin name.

j. Column 10 contains wiring comments. This column is blank or contains an abbreviated comment to identify the function of the signal available at that pin of the device. For example, a pin on a connector will include the term "CONN" to more clearly indicate that this is a connector. On a package with several single output gates (multiple inputs), comments will indicate the pin(s) that correspond to each gate. For example, the pin comments for a common gate will be IN1, IN2, IN3, etc., for the inputs, and OUT1 for the output. Also, abbreviations will be listed for pins on flip-flops and other more complicated devices. Data sheets for the devices reflect these wiring comments.

LOAD LISTING FOR:
PRODUCT CODE 980A

DRAWING NUMBER 960687-9903
REVISION LETTER A

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD		BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
+19.5VA	CB	MEM8KP1	71	CN	0		9999	05P1/604	71	CONN
+19.5VA	CB	MEM8KP2	9	CN	0		9999	05P2/204	9	CONN
+19.5VA	CB	PWRCONN	15	CN	0		9999	09P2/401	15	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
+19.5VB	CB	MEM8KP1	72	CN	0		9999	05P1/604	72	CONN
+19.5VB	CB	MEM8KP2	10	CN	0		9999	05P2/204	10	CONN
+19.5VB	CB	PWRCONN	16	CN	0		9999	09P2/401	16	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
+29VA	CB	PWRCONN	31	CN	0		9999	09P2/401	31	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
+29VB	CB	PWRCONN	32	CN	0		9999	09P2/401	32	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
+35VACA	CB	PWRCONN	21	CN	0		9999	09P2/401	21	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
+35VACB	CB	PWRCONN	22	CN	0		9999	09P2/401	22	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
$00001217	74H61N	AU1C14	8	*	-1		AU1	314	8	EXP2
$00001217	74H61N	AU1C10	8	*	-1		AU1	310	8	EXP2
$00001217	74H61N	AU1C06	8	*	-1		AU1	306	8	EXP2
$00001217	74H52N	AU1M19	9		6		AU1	1219	9	EXP
$00001217	74H61N	AU1C06	10	*	-1		AU1	306	10	EXP3
					2	NET LOADS *MULTIPLE SOURCE*				
$00001222	74H61N	AU1C08	8	*	-1		AU1	308	8	EXP2
$00001222	74H61N	AU1C08	9	*	-1		AU1	308	9	EXP1
$00001222	74H61N	AU1C14	9	*	-1		AU1	314	9	EXP1
$00001222	74H52N	AU1M17	9		6		AU1	1217	9	EXP
$00001222	74H61N	AU1G09	9	*	-1		AU1	709	9	EXP1
					2	NET LOADS *MULTIPLE SOURCE*				
$00001227	74H61N	AU1C12	8	*	-1		AU1	312	8	EXP2
$00001227	74H61N	AU1C07	8	*	-1		AU1	307	8	EXP2
$00001227	74H61N	AU1C07	9	*	-1		AU1	307	9	EXP1
$00001227	74H52N	AU1M18	9		6		AU1	1218	9	EXP
$00001227	74H61N	AU1C07	10	*	-1		AU1	307	10	EXP3
					2	NET LOADS *MULTIPLE SOURCE*				
$00001232	74H61N	AU1B08	8	*	-1		AU1	208	8	EXP2
$00001232	74H52N	AU1B15	9		6		AU1	215	9	EXP
$00001232	74H61N	AU1C08	10	*	-1		AU1	308	10	EXP3
$00001232	74H61N	AU1C10	10	*	-1		AU1	310	10	EXP3
$00001232	74H61N	AU1C12	10	*	-1		AU1	312	10	EXP3
					2	NET LOADS *MULTIPLE SOURCE*				

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
$00001237	74H61N	AU1B14	8	*	-1	AU1	214	8	EXP2
$00001237	74H61N	AU1B13	8	*	-1	AU1	213	8	EXP2
$00001237	74H52N	AU1B18	9		6	AU1	218	9	EXP
$00001237	74H61N	AU1C06	9	*	-1	AU1	306	9	EXP1
$00001237	74H61N	AU1C14	10	*	-1	AU1	314	10	EXP3
					2 NET LOADS *MULTIPLE SOURCE*				
$00001242	74H61N	AU1B12	9	*	-1	AU1	212	9	EXP1
$00001242	74H61N	AU1B14	9	*	-1	AU1	214	9	EXP1
$00001242	74H52N	AU1A15	9		6	AU1	115	9	EXP
$00001242	74H61N	AU1B14	10	*	-1	AU1	214	10	EXP3
$00001242	74H61N	AU1B13	10	*	-1	AU1	213	10	EXP3
					2 NET LOADS *MULTIPLE SOURCE*				
$00001247	74H61N	AU1B08	9	*	-1	AU1	208	9	EXP1
$00001247	74H61N	AU1C12	9	*	-1	AU1	312	9	EXP1
$00001247	74H61N	AU1C10	9	*	-1	AU1	310	9	EXP1
$00001247	74H52N	AU1L19	9		6	AU1	1119	9	EXP
$00001247	74H61N	AU1B08	10	*	-1	AU1	208	10	EXP3
					2 NET LOADS *MULTIPLE SOURCE*				
$00001251	74H61N	AU1B12	8	*	-1	AU1	212	8	EXP2
$00001251	74H61N	AU1B13	9	*	-1	AU1	213	9	EXP1
$00001251	74H52N	AU1A18	9		6	AU1	118	9	EXP
$00001251	74H61N	AU1B12	10	*	-1	AU1	212	10	EXP3
					3 NET LOADS *MULTIPLE SOURCE*				
$00001254	74H55N	AU2R16	9		4	AU2	1516	9	EXP-
$00001254	74H60N	AU2R17	12	*	-1	AU2	1517	12	EXP-
					3 NET LOADS				
$00001255	74H61N	AU1G09	8	*	-1	AU1	709	8	EXP2
$00001255	74H52N	AU1G02	9		6	AU1	702	9	EXP
$00001255	74H61N	AU1G09	10	*	-1	AU1	709	10	EXP3
$00001255	74H61N	AU1N10	10	*	-1	AU1	1310	10	EXP3
					3 NET LOADS * MULTIPLE SOURCE				
$00001256	74H55N	AU2R16	5		4	AU2	1516	5	EXP
$00001256	74H60N	AU2R17	11	*	-1	AU2	1517	11	EXP
					3 NET LOADS				
-35VACA	CB	PWRCONN	19	CN	0	9999	09P2/401	19	CONN
					0 NET LOADS *EXTERNAL SOURCE*				
-35VACB	CB	PWRCONN	20	CN	0	9999	09P2/401	20	CONN
					0 NET LOADS *EXTERNAL SOURCE*				
ADB10-	CB	MEM8KP1	57	CN	0	9999	05P1/604	57	CONN
ADB10-	CB	MEMCONP1	39	CN	0	9999	06P1/605	39	CONN
					0 NET LOADS *EXTERNAL SOURCE*				

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD		BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ADB11-	CB	MEM8KP1	59	CN	0		9999	05P1/604	59	CONN
ADB11-	CB	MEMCONP2	70	CN	0		9999	06P2/205	70	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADB12-	CB	MEM8KP1	61	CN	0		9999	05P1/604	61	CONN
ADB12-	CB	MEMCONP2	71	CN	0		9999	06P2/205	71	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADB13-	CB	MEM8KP1	63	CN	0		9999	05P1/604	63	CONN
ADB13-	CB	MEMCONP2	67	CN	0		9999	06P2/205	67	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADB14-	CB	MEM8KP1	65	CN	0		9999	05P1/604	65	CONN
ADB14-	CB	MEMCONP2	54	CN	0		9999	06P2/205	54	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADB15-	CB	MEM8KP1	67	CN	0		9999	05P1/604	67	CONN
ADB15-	CB	MEMCONP2	60	CN	0		9999	06P2/205	60	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADB6-	CB	MEM8KP1	49	CN	0		9999	05P1/604	49	CONN
ADB6-	CB	MEMCONP2	32	CN	0		9999	06P2/205	32	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADB7-	CB	MEM8KP1	51	CN	0		9999	05P1/604	51	CONN
ADB7-	CB	MEMCONP1	43	CN	0		9999	06P1/605	43	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADB8-	CB	MEM8KP1	53	CN	0		9999	05P1/604	53	CONN
ADB8-	CB	MEMCONP2	23	CN	0		9999	06P2/205	23	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADB9-	CB	MEM8KP1	55	CN	0		9999	05P1/604	55	CONN
ADB9-	CB	MEMCONP2	29	CN	0		9999	06P2/205	29	CONN
					0	NET LOADS *EXTERNAL SOURCE*				
ADDIMM-	7474N	AU2C10	2		-10		AU2	310	2	1D
ADDIMM-	7430N	AU2A08	8	*	100		AU2	108	8	OUT
					90	NET LOADS				
ADRV	74H00N	AU1D18	1		-12		AU1	418	1	INA1
ADRV	7404N	AU1J02	4	*	100		AU1	902	4	OUT2
ADRV	7408N	AU1M04	12		-10		AU1	1204	12	INA4
ADRV	74H52N	AU1P19	12		-12		AU1	1419	12	INH
					66	NET LOADS				
ADRV-	7410N	AU1G16	1		-10		AU1	716	1	INA1
ADRV-	7404N	AU1J02	3		-10		AU1	902	3	IN2

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ADRV-	74H52N	AU1P19	3		-12	AU1	1419	3	INC
ADRV-	7408N	AU2K16	1		-10	AU2	1016	1	INA1
ADRV-	7410N	AU2M12	4		-10	AU2	1212	4	INB2
ADRV-	74H04N	AU2H02	9		-12	AU2	802	9	IN4
ADRV-	CB	AU2P1	46	CN	0	9999	08P1/702	46	CONN
ADRV-	CB	AUXPORT	5	CN	0	9999	10P1/801	5	CONN
ADRV-	CB	AU1P2	62	CN	0	9999	07P2/301	62	CONN
ADRV-	CB	MEMCONP2	73	CN	0	9999	06P2/205	73	CONN
					-64 NET LOADS *EXTERNAL SOURCE*				
AEQB	7400N	AU2M13	4		-10	AU2	1213	4	INA2
AEQB	74H04N	AU2H02	13		-12	AU2	802	13	IN6
AEQB	74181N	AU2G07	14	*	100	AU2	707	14	A=B
AEQB	74181N	AU2J07	14	*	100	AU2	907	14	A=B
AEQB	74181N	AU2L07	14	*	100	AU2	1107	14	A=B
AEQB	RES	AU2B01	14		0	AU2	201	14	OTA*
AEQB	74181N	AU2C07	14	*	100	AU2	307	14	A=B
					378 NET LOADS *MULTIPLE SOURCE*				
AEQB-	74H55N	AU2H12	1		-12	AU2	812	1	INA
AEQB-	74H55N	AU2G12	3		-12	AU2	712	3	INC
AEQB-	74H04N	AU2H02	12	*	120	AU2	802	12	OUT6
					96 NET LOADS				
AEQBX-	7400N	AU2M13	6	*	100	AU2	1213	6	OUT2
AEQBX-	7410N	AU2M12	10		-10	AU2	1212	10	INB3
					90 NET LOADS				
AEQO	74151N	AU2K14	3		10	AU2	1014	3	DIN1
AEQO	74H04N	AU2F14	12	*	120	AU2	614	12	OUT6
					130 NET LOADS				
AEQO-	74H20N	AU2H13	8	*	120	AU2	813	8	OUT2
AEQO-	74H04N	AU2F14	13		-12	AU2	614	13	IN6
AEQO-	74151N	AU2K14	14		10	AU2	1014	14	DIN5
					118 NET LOADS				
AEQZ	74H04N	AU2M21	2	*	120	AU2	1221	2	OUT1
AEQZ	74151N	AU2K14	4		10	AU2	1014	4	DIN0
					130 NET LOADS				
AEQZ-	7410N	AU1N20	3		-10	AU1	1320	3	INA2
AEQZ-	7410N	AU1N18	4		-10	AU1	1318	4	INB2
AEQZ-	74H04N	AU2M21	1		-12	AU2	1221	1	IN1
AEQZ-	74H20N	AU2H13	6	*	120	AU2	813	6	OUT1
AEQZ-	74151N	AU2K14	15		10	AU2	1014	15	DIN4
AEQZ-	CT	AU1CT1	37	CN	0	AU1	07CTP1	37	CONN
AEQZ-	CT	AU2CT1	37	CN	0	AU2	08CTP1	37	CONN
					98 NET LOADS				

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD		BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
AEUZ 0-	7442N	AU1N16	1	*	100		AU1	1316	1	OUT0
					100	NET LOADS *****UNUSED******				
AEUZ 1-	7442N	AU1N16	2	*	100		AU1	1316	2	OUT1
AEUZ 1-	7400N	AU1M15	9		-10		AU1	1215	9	INA3
					90	NET LOADS				
AEUZ 14	7410N	AU1N18	3		-10		AU1	1318	3	INA2
AEUZ 14	7400N	AU1M15	8	*	100		AU1	1215	8	OUT3
					90	NET LOADS				
AEUZ 2-	7442N	AU1N16	3	*	100		AU1	1316	3	OUT2
AEUZ 2-	7400N	AU1N21	4		-10		AU1	1321	4	INA2
					90	NET LOADS				
AEUZ 27	7410N	AU1N20	4		-10		AU1	1320	4	INB2
AEUZ 27	7400N	AU1N21	6	*	100		AU1	1321	6	OUT2
					90	NET LOADS				
AEUZ 3-	7442N	AU1N16	4	*	100		AU1	1316	4	OUT3
					100	NET LOADS *****UNUSED******				
AEUZ 4-	7442N	AU1N16	5	*	100		AU1	1316	5	OUT4
AEUZ 4-	7400N	AU1M15	10		-10		AU1	1215	10	INB3
					90	NET LOADS				
AEUZ 5-	7442N	AU1N16	6	*	100		AU1	1316	6	OUT5
					100	NET LOADS *****UNUSED******				
AEUZ 6	74S65N	AU1K19	5		-10		AU1	1019	5	INH
AEUZ 6	7404N	AU1L16	10	*	100		AU1	1116	10	OUT5
					90	NET LOADS				
AEUZ 6-	7442N	AU1N16	7	*	100		AU1	1316	7	OUT6
AEUZ 6-	7404N	AU1L16	11		-10		AU1	1116	11	IN5
					90	NET LOADS				
AEUZ 7-	7400N	AU1N21	5		-10		AU1	1321	5	INB2
AEUZ 7-	7442N	AU1N16	9	*	100		AU1	1316	9	OUT7
					90	NET LOADS				
AEUZ 8-	7442N	AU1N16	10	*	100		AU1	1316	10	OUT8
					100	NET LOADS *****UNUSED******				
AEUZ 9-	7442N	AU1N16	11	*	100		AU1	1316	11	OUT9
					100	NET LOADS *****UNUSED******				
AGTHN-	74H20N	AU2H16	4		-12		AU2	816	4	INC1

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
AGTHN-	74H55N	AU2G12	8	*	120	AU2	712	8	OUT
					108 NET LOADS				
ALSO	74151N	AU2K14	1		10	AU2	1014	1	DIN3
ALSO	7404N	AU2J14	6	*	100	AU2	914	6	OUT3
					110 NET LOADS				
ALSO-	7404N	AU2J14	5		-10	AU2	914	5	IN3
ALSO-	74151N	AU2K13	6	*	100	AU2	1013	6	OUT-
ALSO-	74S00N	AU2E10	9		-10	AU2	510	9	INA3
ALSO-	74H21N	AU2E11	9		-12	AU2	511	9	INA2
ALSO-	74151N	AU2K14	12		10	AU2	1014	12	DIN7
					78 NET LOADS				
ALS1-	74151N	AU2J12	6	*	100	AU2	912	6	OUT-
ALS1-	74H21N	AU2E11	10		-12	AU2	511	10	INB2
ALS1-	74S00N	AU2E10	12		-10	AU2	510	12	INA4
					78 NET LOADS				
ALS10-	74151N	AU2P09	6	*	100	AU2	1409	6	OUT-
ALS10-	74S00N	AU2H11	9		-10	AU2	811	9	INA3
ALS10-	74H21N	AU2H08	12		-12	AU2	808	12	INC2
					78 NET LOADS				
ALS11-	74S00N	AU2J06	4		-10	AU2	906	4	INA2
ALS11-	74151N	AU2N08	6	*	100	AU2	1308	6	OUT-
ALS11-	74H21N	AU2H08	13		-12	AU2	808	13	IND2
					78 NET LOADS				
ALS12-	74S00N	AU2L06	1		-10	AU2	1106	1	INA1
ALS12-	74151N	AU2R11	6	*	100	AU2	1511	6	OUT-
ALS12-	74H21N	AU2K07	9		-12	AU2	1007	9	INA2
					78 NET LOADS				
ALS13-	74151N	AU2R08	6	*	100	AU2	1508	6	OUT-
ALS13-	74S00N	AU2J06	9		-10	AU2	906	9	INA3
ALS13-	74H21N	AU2K07	10		-12	AU2	1007	10	INB2
					78 NET LOADS				
ALS14-	74151N	AU2R10	6	*	100	AU2	1510	6	OUT-
ALS14-	74S00N	AU2L06	12		-10	AU2	1106	12	INA4
ALS14-	74H21N	AU2K07	12		-12	AU2	1007	12	INC2
					78 NET LOADS				
ALS15-	74151N	AU2R13	6	*	100	AU2	1513	6	OUT-
ALS15-	74S00N	AU2L06	9		-10	AU2	1106	9	INA3
ALS15-	74151N	AU2K14	13		10	AU2	1014	13	DIN6
ALS15-	74H21N	AU2K07	13		-12	AU2	1007	13	IND2
					88 NET LOADS				

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ALS2-	74S00N	AU2E10	4		-10	AU2	510	4	INA2
ALS2-	74151N	AU2M11	6	*	100	AU2	1211	6	OUT-
ALS2-	74H21N	AU2E11	12		-12	AU2	511	12	INC2
					78 NET LOADS				
ALS3-	74S0CN	AU2E10	1		-10	AU2	510	1	INA1
ALS3-	74151N	AU2L12	6	*	100	AU2	1112	6	OUT-
ALS3-	74H21N	AU2E11	13		-12	AU2	511	13	IND2
					78 NET LOADS				
ALS4-	74S0CN	AU2G11	1		-10	AU2	711	1	INA1
ALS4-	74151N	AU2L10	6	*	100	AU2	1110	6	OUT-
ALS4-	74H21N	AU2H10	9		-12	AU2	810	9	INA2
					78 NET LOADS				
ALS5-	74S00N	AU2G11	4		-10	AU2	711	4	INA2
ALS5-	74151N	AU2K11	6	*	100	AU2	1011	6	OUT-
ALS5-	74H21N	AU2H10	10		-12	AU2	810	10	INB2
					78 NET LOADS				
ALS6-	74S0CN	AU2H11	1		-10	AU2	811	1	INA1
ALS6-	74151N	AU2M09	6	*	100	AU2	1209	6	OUT-
ALS6-	74H21N	AU2H10	12		-12	AU2	810	12	INC2
					78 NET LOADS				
ALS7-	74S00N	AU2H11	4		-10	AU2	811	4	INA2
ALS7-	74151N	AU2K08	6	*	100	AU2	1008	6	OUT-
ALS7-	74H21N	AU2H10	13		-12	AU2	810	13	IND2
					78 NET LOADS				
ALS8-	74S00N	AU2J06	1		-10	AU2	906	1	INA1
ALS8-	74151N	AU2P07	6	*	100	AU2	1407	6	OUT-
ALS8-	74H21N	AU2H08	9		-12	AU2	808	9	INA2
					78 NET LOADS				
ALS9-	74151N	AU2P08	6	*	100	AU2	1408	6	OUT-
ALS9-	74H21N	AU2H08	10		-12	AU2	808	10	INB2
ALS9-	74S0CN	AU2J06	12		-10	AU2	906	12	INA4
					78 NET LOADS				
ALUB0	74H11N	AU2C03	6	*	120	AU2	303	6	OUT2
ALUB0	CB	MEMCONP1	54	CN	0	9999	06P1/605	54	CONN
ALUB0	CB	AU2P2	30	CN	0	9999	08P2/302	30	CONN
					120 NET LOADS				
ALUB1	74H11N	AU2A02	12	*	120	AU2	102	12	OUT1
ALUB1	CB	MEMCONP1	56	CN	0	9999	06P1/605	56	CONN
ALUB1	CB	AU2P2	22	CN	0	9999	08P2/302	22	CONN

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
					120 NET LOADS				
ALUB10	74H11N	AU2F03	6	*	120	AU2	603	6	OUT2
ALUB10	CB	AU2P2	71	CN	0	9999	08P2/302	71	CONN
ALUB10	CB	MEMCONP2	75	CN	0	9999	06P2/205	75	CONN
					120 NET LOADS				
ALUB11	74H11N	AU2C02	8	*	120	AU2	302	8	OUT3
ALUB11	CB	MEMCONP1	26	CN	0	9999	06P1/605	26	CONN
ALUB11	CB	AU2P2	29	CN	0	9999	08P2/302	29	CONN
					120 NET LOADS				
ALUB12	74H11N	AU2F02	6	*	120	AU2	602	6	OUT2
ALUB12	CB	MEMCONP1	6	CN	0	9999	06P1/605	6	CONN
ALUB12	CB	AU2P2	73	CN	0	9999	08P2/302	73	CONN
					120 NET LOADS				
ALUB13	74H11N	AU2F03	12	*	120	AU2	603	12	OUT1
ALUB13	CB	MEMCONP1	46	CN	0	9999	06P1/605	46	CONN
ALUB13	CB	AU2P2	69	CN	0	9999	08P2/302	69	CONN
					120 NET LOADS				
ALUB14	74H11N	AU2G02	6	*	120	AU2	702	6	OUT2
ALUB14	CB	MEMCONP1	10	CN	0	9999	06P1/605	10	CONN
ALUB14	CB	AU2P2	63	CN	0	9999	08P2/302	63	CONN
					120 NET LOADS				
ALUB15	74H11N	AU2C02	6	*	120	AU2	302	6	OUT2
ALUB15	CB	MEMCONP1	9	CN	0	9999	06P1/605	9	CONN
ALUB15	CB	AU2P2	31	CN	0	9999	08P2/302	31	CONN
					120 NET LOADS				
ALUB2	74H11N	AU2A02	8	*	120	AU2	102	8	OUT3
ALUB2	CB	MEMCONP1	60	CN	0	9999	06P1/605	60	CONN
ALUB2	CB	AU2P2	33	CN	0	9999	08P2/302	33	CONN
					120 NET LOADS				
ALUB3	74H11N	AU2A02	6	*	120	AU2	102	6	OUT2
ALUB3	CB	MEMCONP1	52	CN	0	9999	06P1/605	52	CONN
ALUB3	CB	AU2P2	20	CN	0	9999	08P2/302	20	CONN
					120 NET LOADS				
ALUB4	74H11N	AU2C02	12	*	120	AU2	302	12	OUT1
ALUB4	CB	MEMCONP1	65	CN	0	9999	06P1/605	65	CONN
ALUB4	CB	AU2P2	25	CN	0	9999	08P2/302	25	CONN
					120 NET LOADS				
ALUB5	74H11N	AU2F02	12	*	120	AU2	602	12	OUT1
ALUB5	CB	MEMCONP1	70	CN	0	9999	06P1/605	70	CONN

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ALUB5	CB	AU2P2	67	CN	0	9999	08P2/302	67	CONN
					120 NET LOADS				
ALUB6	74H11N	AU2E03	6	*	120	AU2	503	6	OUT2
ALUB6	CB	MEMCONP1	68	CN	0	9999	06P1/605	68	CONN
ALUB6	CB	AU2P2	59	CN	0	9999	08P2/302	59	CONN
					120 NET LOADS				
ALUB7	74H11N	AU2F02	8	*	120	AU2	602	8	OUT3
ALUB7	CB	MEMCONP1	48	CN	0	9999	06P1/605	48	CONN
ALUB7	CB	AU2P2	74	CN	0	9999	08P2/302	74	CONN
					120 NET LOADS				
ALUB8	74H11N	AU2E03	12	*	120	AU2	503	12	OUT1
ALUB8	CB	MEMCONP1	22	CN	0	9999	06P1/605	22	CONN
ALUB8	CB	AU2P2	54	CN	0	9999	08P2/302	54	CONN
					120 NET LOADS				
ALUB9	74H11N	AU2E03	8	*	120	AU2	503	8	OUT3
ALUB9	CB	MEMCONP1	30	CN	0	9999	06P1/605	30	CONN
ALUB9	CB	AU2P2	61	CN	0	9999	08P2/302	61	CONN
					120 NET LOADS				
ALUEQA-	7410N	AU1D20	6	*	100	AU1	420	6	OUT2
ALUEQA-	7437N	AU2F16	5		-10	AU2	616	5	INB2
ALUEQA-	7437N	AU2F16	10		-10	AU2	616	10	INB3
ALUEQA-	CT	AU1CT2	42	CN	0	AU1	07CTP2	42	CONN
ALUEQA-	CT	AU2CT2	42	CN	0	AU2	08CTP2	42	CONN
					80 NET LOADS				
ALUEQURO	7486N	AU2F19	3	*	100	AU2	619	3	OUT1
ALUEQURO	74H52N	AU2F20	12		-12	AU2	620	12	INH
					88 NET LOADS				
ALUEQZ	7410N	AU1G16	5		-10	AU1	716	5	INC2
ALUEQZ	74S65N	AU1K19	6		-10	AU1	1019	6	INI
ALUEQZ	74S65N	AU1K19	10		-10	AU1	1019	10	INK
ALUEQZ	74H60N	AU2R17	1		-12	AU2	1517	1	INA1
ALUEQZ	74S11N	AU2K18	8	*	100	AU2	1018	8	OUT3
ALUEQZ	CT	AU1CT1	50	CN	0	AU1	07CTP1	50	CONN
ALUEQZ	CT	AU2CT1	50	CN	0	AU2	08CTP1	50	CONN
					58 NET LOADS				
ALUEQ8K	74S04N	AU1E08	2	*	100	AU1	508	2	OUT1
ALUEQ8K	74S03N	AU1D06	9		-10	AU1	406	9	INA3
					90 NET LOADS				
ALUEQ8K-	74S04N	AU1E08	1		-10	AU1	508	1	IN1
ALUEQ8K-	74S65N	AU1K19	3		-10	AU1	1019	3	INF

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ALUEQ8K-	74S22N	AU1C21	9		-10	AU1	321	9	INA2
ALUEQ8K-	74H10N	AU1M16	13		-12	AU1	1216	13	INC1
ALUEQ8K-	74H30N	AU2L15	4		-12	AU2	1115	4	IND
ALUEQ8K-	74S10N	AU2D21	12	*	100	AU2	421	12	OUT1
ALUEQ8K-	CT	AU1CT2	60	CN	0	AU1	07CTP2	60	CONN
ALUEQ8K-	CT	AU2CT2	60	CN	0	AU2	08CTP2	60	CONN
					46 NET LOADS				
ALUM	7404N	AU2J14	4	*	100	AU2	914	4	OUT2
ALUM	74181N	AU2J07	8		-10	AU2	907	8	INM
ALUM	74181N	AU2G07	8		-10	AU2	707	8	INM
ALUM	74181N	AU2L07	8		-10	AU2	1107	8	INM
ALUM	74181N	AU2C07	8		-10	AU2	307	8	INM
					60 NET LOADS				
ALUS0	74181N	AU2C07	6		-40	AU2	307	6	INS0
ALUS0	74181N	AU2L07	6		-40	AU2	1107	6	INS0
ALUS0	74181N	AU2G07	6		-40	AU2	707	6	INS0
ALUS0	74181N	AU2J07	6		-40	AU2	907	6	INS0
ALUS0	7437N	AU2F16	8	*	300	AU2	616	8	OUT3
					140 NET LOADS				
ALUS1	74181N	AU2J07	5		-40	AU2	907	5	INS1
ALUS1	74181N	AU2G07	5		-40	AU2	707	5	INS1
ALUS1	74181N	AU2L07	5		-40	AU2	1107	5	INS1
ALUS1	74181N	AU2C07	5		-40	AU2	307	5	INS1
ALUS1	7437N	AU2B08	11	*	300	AU2	208	11	OUT4
					140 NET LOADS				
ALUS2	74181N	AU2C07	4		-40	AU2	307	4	INS2
ALUS2	74181N	AU2L07	4		-40	AU2	1107	4	INS2
ALUS2	74181N	AU2G07	4		-40	AU2	707	4	INS2
ALUS2	74181N	AU2J07	4		-40	AU2	907	4	INS2
ALUS2	7437N	AU2F16	6	*	300	AU2	616	6	OUT2
ALUS2	74H51N	AU2D14	13		-12	AU2	414	13	INB1
					128 NET LOADS				
ALUS3	74181N	AU2J07	3		-50	AU2	907	3	INS3
ALUS3	74181N	AU2G07	3		-50	AU2	707	3	INS3
ALUS3	74181N	AU2L07	3		-50	AU2	1107	3	INS3
ALUS3	74181N	AU2C07	3		-50	AU2	307	3	INS3
ALUS3	7437N	AU2B08	3	*	300	AU2	208	3	OUT1
					100 NET LOADS				
ALUZ115	74S65N	AU2B07	8	*	100	AU2	207	8	OUT1
ALUZ115	74S65N	AU2N03	8	*	100	AU2	1303	8	OUT1
ALUZ115	74S65N	AU2R03	8	*	100	AU2	1503	8	OUT1
ALUZ115	74S65N	AU2P03	8	*	100	AU2	1403	8	OUT1
ALUZ115	RES	AU2B10	9		0	AU2	210	9	OTA*

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ALUZ115	74S11N	AU2K18	11		-10	AU2	1018	11	INC3
ALUZ115	74S10N	AU2D21	13		-10	AU2	421	13	INC1
					380 NET LOADS *MULTIPLE SOURCE*				
ALUO	74S10N	AU2D21	2		-10	AU2	421	2	INB1
ALUO	74157N	AU2G03	3		-10	AU2	703	3	INB1
ALUO	74H11N	AU2C03	3		-12	AU2	303	3	INA2
ALUO	74153N	AU2C12	6		-10	AU2	312	6	1CO
ALUO	74153N	AU2F05	6		-10	AU2	605	6	1CO
ALUO	74H04N	AU2E04	11		-12	AU2	504	11	IN5
ALUO	74S00N	AU2J19	12		-10	AU2	919	12	INA4
ALUO	74153N	AU2D13	12		-10	AU2	413	12	2C2
ALUO	74181N	AU2C07	13	*	100	AU2	307	13	F3
					16 NET LOADS				
ALUO-	74H55N	AU2L14	1		-12	AU2	1114	1	INA
ALUO-	7410N	AU2C18	5		-10	AU2	318	5	INC2
ALUO-	74S11N	AU2K18	10		-10	AU2	1018	10	INB3
ALUO-	74H51N	AU2D18	10		-12	AU2	418	10	IND1
ALUO-	74H04N	AU2E04	10	*	120	AU2	504	10	OUT5
ALUO-	74H55N	AU2R16	11		-12	AU2	1516	11	INF
					64 NET LOADS				
ALUOA-	74S40N	AU2K19	9		-10	AU2	1019	9	INA2
ALUOA-	74S00N	AU2J19	11	*	120	AU2	919	11	DOUT
					110 NET LOADS				
ALUOT	74H11N	AU1H21	3		-12	AU1	821	3	INA2
ALUOT	74S65N	AU1J17	4		-10	AU1	917	4	ING
ALUOT	74S22N	AU1H19	4		-10	AU1	819	4	INC1
ALUOT	74S22N	AU1K18	4		-10	AU1	1018	4	INC1
ALUOT	74S10N	AU1E20	9		-10	AU1	520	9	INA3
ALUOT	74S22N	AU1K18	9		-10	AU1	1018	9	INA2
ALUOT	74S04N	AU1E08	9		-10	AU1	508	9	IN4
ALUOT	7486N	AU2F19	1		-10	AU2	619	1	INA1
ALUOT	74H51N	AU2D18	1		-12	AU2	418	1	INA1
ALUOT	74H55N	AU2G12	2		-12	AU2	712	2	INB
ALUOT	74H55N	AU2H12	2		-12	AU2	812	2	INB
ALUOT	74H52N	AU2F20	2		-12	AU2	620	2	INB
ALUOT	74H55N	AU2R16	2		-12	AU2	1516	2	INB
ALUOT	74H10N	AU2F15	4		-12	AU2	615	4	INB2
ALUOT	74H51N	AU2D14	5		-12	AU2	414	5	IND2
ALUOT	74H52N	AU2J15	5		-12	AU2	915	5	INE
ALUOT	74S40N	AU2K19	8	*	300	AU2	1019	8	OUT2
ALUOT	7400N	AU2N15	12		-10	AU2	1315	12	INA4
ALUOT	74H55N	AU2L14	13		-12	AU2	1114	13	INH
ALUOT	CT	AU1CT1	54	CN	0	AU1	07CTP1	54	CONN
ALUOT	CT	AU2CT1	54	CN	0	AU2	08CTP1	54	CONN
					100 NET LOADS				

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ALUOT-	74S22N	AU1C21	1		-10	AU1	321	1	INA1
ALUOT-	74S65N	AU1J17	1		-10	AU1	917	1	INA
ALUOT-	74S22N	AU1H18	4		-10	AU1	818	4	INC1
ALUOT-	74S04N	AU1E08	8	*	100	AU1	508	8	OUT4
ALUOT-	74S22N	AU1H18	12		-10	AU1	818	12	INC2
					60 NET LOADS				
ALU1	74H11N	AU2A02	1		-12	AU2	102	1	INA1
ALU1	74S65N	AU2B07	2		-10	AU2	207	2	INE
ALU1	74153N	AU2P15	3		-10	AU2	1415	3	1C3
ALU1	74H51N	AU2D14	3		-12	AU2	414	3	INB2
ALU1	74153N	AU2D05	4		-10	AU2	405	4	1C2
ALU1	74157N	AU2G03	6		-10	AU2	703	6	INB2
ALU1	7400N	AU2M13	9		-10	AU2	1213	9	INA3
ALU1	74153N	AU2F05	10		-10	AU2	605	10	2C0
ALU1	74181N	AU2C07	11	*	100	AU2	307	11	F2
ALU1	74153N	AU2D13	13		-10	AU2	413	13	2C3
					6 NET LOADS				
ALU10	74153N	AU2F11	3		-10	AU2	611	3	1C3
ALU10	74H11N	AU2F03	3		-12	AU2	603	3	INA2
ALU10	74194N	AU2N11	5		-10	AU2	1311	5	INC
ALU10	74153N	AU2D05	6		-10	AU2	405	6	1C0
ALU10	74S65N	AU2P03	9		-10	AU2	1403	9	INJ
ALU10	74181N	AU2J07	10	*	100	AU2	907	10	F1
ALU10	74153N	AU2F05	11		-10	AU2	605	11	2C1
ALU10	74153N	AU2F11	12		-10	AU2	611	12	2C2
ALU10	7474N	AU2P19	12		-10	AU2	1419	12	2D
ALU10	74157N	AU2P02	13		-10	AU2	1402	13	INB3
					8 NET LOADS				
ALU11	74S65N	AU2P03	1		-10	AU2	1403	1	INA
ALU11	74153N	AU2F08	4		-10	AU2	608	4	1C2
ALU11	74153N	AU2F11	5		-10	AU2	611	5	1C1
ALU11	74194N	AU2N11	6		-10	AU2	1311	6	IND
ALU11	74181N	AU2J07	9	*	100	AU2	907	9	F0
ALU11	74H11N	AU2C02	9		-12	AU2	302	9	INA3
ALU11	74153N	AU2D05	10		-10	AU2	405	10	2C0
ALU11	74157N	AU2P02	10		-10	AU2	1402	10	INB4
ALU11	7408N	AU2N13	12		-10	AU2	1313	12	INA4
ALU11	74153N	AU2F11	13		-10	AU2	611	13	2C3
					8 NET LOADS				
ALU12	74153N	AU2F08	3		-10	AU2	608	3	1C3
ALU12	74H11N	AU2F02	3		-12	AU2	602	3	INA2
ALU12	74194N	AU2L17	3		-10	AU2	1117	3	INA
ALU12	74157N	AU2P01	3		-10	AU2	1401	3	INB1
ALU12	74S65N	AU2P03	4		-10	AU2	1403	4	ING

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ALU12	74153N	AU2E08	6		-10	AU2	508	6	1C0
ALU12	7408N	AU2N13	9		-10	AU2	1313	9	INA3
ALU12	74153N	AU2F11	11		-10	AU2	611	11	2C1
ALU12	74153N	AU2F08	12		-10	AU2	608	12	2C2
ALU12	74181N	AU2L07	13	*	100	AU2	1107	13	F3
					8 NET LOADS				
ALU13	74H11N	AU2F03	1		-12	AU2	603	1	INA1
ALU13	74S65N	AU2R03	2		-10	AU2	1503	2	INE
ALU13	74194N	AU2L17	4		-10	AU2	1117	4	INB
ALU13	74153N	AU2E12	4		-10	AU2	512	4	1C2
ALU13	74153N	AU2F08	5		-10	AU2	608	5	1C1
ALU13	74157N	AU2P01	6		-10	AU2	1401	6	INB2
ALU13	74153N	AU2E08	10		-10	AU2	508	10	2C0
ALU13	74181N	AU2L07	11	*	100	AU2	1107	11	F2
ALU13	74153N	AU2F08	13		-10	AU2	608	13	2C3
					18 NET LOADS				
ALU14	74153N	AU2E12	3		-10	AU2	512	3	1C3
ALU14	74H11N	AU2G02	3		-12	AU2	702	3	INA2
ALU14	74194N	AU2L17	5		-10	AU2	1117	5	INC
ALU14	74153N	AU2E05	6		-10	AU2	505	6	1C0
ALU14	74S65N	AU2R03	9		-10	AU2	1503	9	INJ
ALU14	74181N	AU2L07	10	*	100	AU2	1107	10	F1
ALU14	74153N	AU2F08	11		-10	AU2	608	11	2C1
ALU14	74H52N	AU2J15	11		-12	AU2	915	11	ING
ALU14	74153N	AU2E12	12		-10	AU2	512	12	2C2
ALU14	74157N	AU2P01	13		-10	AU2	1401	13	INB3
					6 NET LOADS				
ALU15	74S65N	AU2R03	1		-10	AU2	1503	1	INA
ALU15	74H11N	AU2C02	3		-12	AU2	302	3	INA2
ALU15	74153N	AU2C12	4		-10	AU2	312	4	1C2
ALU15	74S11N	AU2K18	5		-10	AU2	1018	5	INC2
ALU15	74153N	AU2E12	5		-10	AU2	512	5	1C1
ALU15	74194N	AU2L17	6		-10	AU2	1117	6	IND
ALU15	74181N	AU2L07	9	*	100	AU2	1107	9	F0
ALU15	74157N	AU2P01	10		-10	AU2	1401	10	INB4
ALU15	74153N	AU2E05	10		-10	AU2	505	10	2C0
ALU15	74H04N	AU2E04	13		-12	AU2	504	13	IN6
					6 NET LOADS				
ALU15-	74H51N	AU2E16	3		-12	AU2	516	3	INB2
ALU15-	74H04N	AU2E04	12	*	120	AU2	504	12	OUT5
					108 NET LOADS				
ALU15A	74H51N	AU2E16	5		-12	AU2	516	5	IND2
ALU15A	74S11N	AU2K18	6	*	100	AU2	1018	6	OUT2
ALU15A	74H52N	AU2J15	13		-12	AU2	915	13	INI

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
					76 NET LOADS				
ALU2	7400N	AU2N15	1		-10	AU2	1315	1	INA1
ALU2	74153N	AU2D05	3		-10	AU2	405	3	1C3
ALU2	74153N	AU2F11	6		-10	AU2	611	6	1C0
ALU2	74S65N	AU2B07	9		-10	AU2	207	9	INJ
ALU2	74H11N	AU2A02	9		-12	AU2	102	9	INA3
ALU2	74181N	AU2C07	10	*	100	AU2	307	10	F1
ALU2	74153N	AU2D13	11		-10	AU2	413	11	2C1
ALU2	74153N	AU2D05	12		-10	AU2	405	12	2C2
ALU2	74157N	AU2G03	13		-10	AU2	703	13	INB3
ALU2	74153N	AU2N14	13		-10	AU2	1314	13	2C3
					8 NET LOADS				
ALU3	74S65N	AU2B07	1		-10	AU2	207	1	INA
ALU3	74H11N	AU2A02	3		-12	AU2	102	3	INA2
ALU3	74153N	AU2N14	3		-10	AU2	1314	3	1C3
ALU3	74153N	AU2E08	4		-10	AU2	508	4	1C2
ALU3	74153N	AU2D05	5		-10	AU2	405	5	1C1
ALU3	74181N	AU2C07	9	*	100	AU2	307	9	F0
ALU3	74157N	AU2G03	10		-10	AU2	703	10	INB4
ALU3	74153N	AU2F11	10		-10	AU2	611	10	2C0
ALU3	7400N	AU2H14	13		-10	AU2	814	13	INB4
ALU3	74153N	AU2D05	13		-10	AU2	405	13	2C3
					8 NET LOADS				
ALU4	74H11N	AU2C02	1		-12	AU2	302	1	INA1
ALU4	7403N	AU2J13	1		-10	AU2	913	1	INA1
ALU4	74153N	AU2E08	3		-10	AU2	508	3	1C3
ALU4	74157N	AU2J02	3		-10	AU2	902	3	INB1
ALU4	74194N	AU2M10	3		-10	AU2	1210	3	INA
ALU4	74S65N	AU2B07	4		-10	AU2	207	4	ING
ALU4	74153N	AU2F08	6		-10	AU2	608	6	1C0
ALU4	74153N	AU2D05	11		-10	AU2	405	11	2C1
ALU4	74153N	AU2E08	12		-10	AU2	508	12	2C2
ALU4	74181N	AU2G07	13	*	100	AU2	707	13	F3
					8 NET LOADS				
ALU5	74H11N	AU2F02	1		-12	AU2	602	1	INA1
ALU5	74S65N	AU2N03	2		-10	AU2	1303	2	INE
ALU5	74194N	AU2M10	4		-10	AU2	1210	4	INB
ALU5	74153N	AU2E05	4		-10	AU2	505	4	1C2
ALU5	74153N	AU2E08	5		-10	AU2	508	5	1C1
ALU5	74157N	AU2J02	6		-10	AU2	902	6	INB2
ALU5	74153N	AU2F08	10		-10	AU2	608	10	2C0
ALU5	74181N	AU2G07	11	*	100	AU2	707	11	F2
ALU5	74153N	AU2E08	13		-10	AU2	508	13	2C3
					18 NET LOADS				

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
ALU6	74H11N	AU2E03	3		-12	AU2	503	3	INA2
ALU6	74153N	AU2E05	3		-10	AU2	505	3	1C3
ALU6	74194N	AU2M10	5		-10	AU2	1210	5	INC
ALU6	74153N	AU2E12	6		-10	AU2	512	6	1C0
ALU6	74S65N	AU2N03	9		-10	AU2	1303	9	INJ
ALU6	74181N	AU2G07	10	*	100	AU2	707	10	F1
ALU6	74153N	AU2E08	11		-10	AU2	508	11	2C1
ALU6	74153N	AU2E05	12		-10	AU2	505	12	2C2
ALU6	74157N	AU2J02	13		-10	AU2	902	13	INB3
					18 NET LOADS				
ALU7	74S65N	AU2N03	1		-10	AU2	1303	1	INA
ALU7	7408N	AU2N13	1		-10	AU2	1313	1	INA1
ALU7	74153N	AU2F05	4		-10	AU2	605	4	1C2
ALU7	74153N	AU2E05	5		-10	AU2	505	5	1C1
ALU7	74194N	AU2M10	6		-10	AU2	1210	6	IND
ALU7	74H11N	AU2F02	9		-12	AU2	602	9	INA3
ALU7	74181N	AU2G07	9	*	100	AU2	707	9	F0
ALU7	74153N	AU2E12	10		-10	AU2	512	10	2C0
ALU7	74157N	AU2J02	10		-10	AU2	902	10	INB4
ALU7	74153N	AU2E05	13		-10	AU2	505	13	2C3
					8 NET LOADS				
ALU8	74H11N	AU2E03	1		-12	AU2	503	1	INA1
ALU8	74153N	AU2F05	3		-10	AU2	605	3	1C3
ALU8	74157N	AU2P02	3		-10	AU2	1402	3	INB1
ALU8	74194N	AU2N11	3		-10	AU2	1311	3	INA
ALU8	74S65N	AU2N03	4		-10	AU2	1303	4	ING
ALU8	7408N	AU2N13	4		-10	AU2	1313	4	INA2
ALU8	74153N	AU2D13	6		-10	AU2	413	6	1C0
ALU8	74153N	AU2E05	11		-10	AU2	505	11	2C1
ALU8	74153N	AU2F05	12		-10	AU2	605	12	2C2
ALU8	74181N	AU2J07	13	*	100	AU2	907	13	F3
					8 NET LOADS				
ALU9	74S65N	AU2P03	2		-10	AU2	1403	2	INE
ALU9	74194N	AU2N11	4		-10	AU2	1311	4	INB
ALU9	74153N	AU2F11	4		-10	AU2	611	4	1C2
ALU9	74153N	AU2F05	5		-10	AU2	605	5	1C1
ALU9	74157N	AU2P02	6		-10	AU2	1402	6	INB2
ALU9	74C8N	AU2J13	9		-10	AU2	913	9	INA3
ALU9	74H11N	AU2E03	9		-12	AU2	503	9	INA3
ALU9	74153N	AU2D13	10		-10	AU2	413	10	2C0
ALU9	74181N	AU2J07	11	*	100	AU2	907	11	F2
ALU9	74153N	AU2F05	13		-10	AU2	605	13	2C3
					8 NET LOADS				
ALOE1U0	74H11N	AU1H21	6	*	120	AU1	821	6	OUT2
ALOE1U0	74S65N	AU1K19	11		-10	AU1	1019	11	INB

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
					110 NET LOADS				
ALOE1UO-	74S10N	AU1E20	1		-10	AU1	520	1	INA1
ALOE1UO-	74S10N	AU1E20	8	*	100	AU1	520	8	OUT3
					90 NET LOADS				
ALOTOCO-	74H51N	AU2D14	2		-12	AU2	414	2	INA2
ALOTOCO-	74H04N	AU2C13	10	*	120	AU2	313	10	OUT5
					108 NET LOADS				
AMADOA-	74S11N	AU2K18	1		-10	AU2	1018	1	INA1
AMADOA-	74151N	AU2F09	7		10	AU2	609	7	STB
AMADOA-	74151N	AU2F10	7		10	AU2	610	7	STB
AMADOA-	74151N	AU2D08	7		10	AU2	408	7	STB
AMADOA-	74151N	AU2D10	7		10	AU2	410	7	STB
AMADOA-	74151N	AU2D12	7		10	AU2	412	7	STB
AMADOA-	74151N	AU2D09	7		10	AU2	409	7	STB
AMADOA-	74151N	AU2D11	7		10	AU2	411	7	STB
AMADOA-	74S64N	AU2E18	8	*	100	AU2	518	8	OUT1
AMADOA-	74S10N	AU2D21	9		-10	AU2	421	9	INA3
AMADOA-	74S20N	AU2E21	9		-10	AU2	521	9	INA2
					140 NET LOADS				
AMADOB-	74151N	AU2J10	7		10	AU2	910	7	STB
AMADOB-	74151N	AU2J03	7		10	AU2	903	7	STB
AMADOB-	74151N	AU2J04	7		10	AU2	904	7	STB
AMADOB-	74151N	AU2J05	7		10	AU2	905	7	STB
AMADOB-	74151N	AU2K04	7		10	AU2	1004	7	STB
AMADOB-	74151N	AU2L05	7		10	AU2	1105	7	STB
AMADOB-	74151N	AU2N04	7		10	AU2	1304	7	STB
AMADOB-	74151N	AU2M03	7		10	AU2	1203	7	STB
AMADOB-	74151N	AU2L04	7		10	AU2	1104	7	STB
AMADOB-	74S11N	AU2K18	12	*	100	AU2	1018	12	OUT1
					190 NET LOADS				
AMAD1	74S40N	AU2K19	6	*	300	AU2	1019	6	OUT1
AMAD1	74151N	AU2P09	9		10	AU2	1409	9	DS C
AMAD1	74151N	AU2M11	9		10	AU2	1211	9	DS C
AMAD1	74151N	AU2R10	9		10	AU2	1510	9	DS C
AMAD1	74151N	AU2R11	9		10	AU2	1511	9	DS C
AMAD1	74151N	AU2N08	9		10	AU2	1308	9	DS C
AMAD1	74151N	AU2R08	9		10	AU2	1508	9	DS C
AMAD1	74151N	AU2P08	9		10	AU2	1408	9	DS C
AMAD1	74151N	AU2P07	9		10	AU2	1407	9	DS C
AMAD1	74151N	AU2M03	9		10	AU2	1203	9	DS C
AMAD1	74151N	AU2L04	9		10	AU2	1104	9	DS C
AMAD1	74151N	AU2L10	9		10	AU2	1110	9	DS C
AMAD1	74151N	AU2N04	9		10	AU2	1304	9	DS C
AMAD1	74151N	AU2M09	9		10	AU2	1209	9	DS C

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
AMAD1	74151N	AU2L05	9		10	AU2	1105	9	DS C
AMAD1	74151N	AU2R13	9		10	AU2	1513	9	DS C
AMAD1	74151N	AU2L12	9		10	AU2	1112	9	DS C
AMAD1	74151N	AU2K04	9		10	AU2	1004	9	DS C
AMAD1	74151N	AU2J10	9		10	AU2	910	9	DS C
AMAD1	74151N	AU2J05	9		10	AU2	905	9	DS C
AMAD1	74151N	AU2J12	9		10	AU2	912	9	DS C
AMAD1	74151N	AU2J03	9		10	AU2	903	9	DS C
AMAD1	74151N	AU2F10	9		10	AU2	610	9	DS C
AMAD1	74151N	AU2K13	9		10	AU2	1013	9	DS C
AMAD1	74151N	AU2F09	9		10	AU2	609	9	DS C
AMAD1	74151N	AU2K08	9		10	AU2	1008	9	DS C
AMAD1	74151N	AU2K11	9		10	AU2	1011	9	DS C
AMAD1	74151N	AU2J04	9		10	AU2	904	9	DS C
AMAD1	74151N	AU2D11	9		10	AU2	411	9	DS C
AMAD1	74151N	AU2D09	9		10	AU2	409	9	DS C
AMAD1	74151N	AU2D08	9		10	AU2	408	9	DS C
AMAD1	74151N	AU2D12	9		10	AU2	412	9	DS C
AMAD1	74151N	AU2D10	9		10	AU2	410	9	DS C
					620 NET LOADS				
AMAD1A-	74S40N	AU2K19	1		-10	AU2	1019	1	INA1
AMAD1A-	74S00N	AU2J19	6	*	120	AU2	919	6	BOUT
					110 NET LOADS				
AMAD1B-	74S40N	AU2K19	2		-10	AU2	1019	2	INB1
AMAD1B-	74S00N	AU2J19	8	*	120	AU2	919	8	COUT
					110 NET LOADS				
AMAD1C-	74S40N	AU2K19	4		-10	AU2	1019	4	INC1
AMAD1C-	74S00N	AU2H21	8	*	120	AU2	821	8	COUT
					110 NET LOADS				
AMAD2	74S40N	AU2G14	8	*	300	AU2	714	8	OUT2
AMAD2	74151N	AU2J10	10		10	AU2	910	10	DS B
AMAD2	74151N	AU2J05	10		10	AU2	905	10	DS B
AMAD2	74151N	AU2J12	10		10	AU2	912	10	DS B
AMAD2	74151N	AU2J04	10		10	AU2	904	10	DS B
AMAD2	74151N	AU2K11	10		10	AU2	1011	10	DS B
AMAD2	74151N	AU2K04	10		10	AU2	1004	10	DS B
AMAD2	74151N	AU2F09	10		10	AU2	609	10	DS B
AMAD2	74151N	AU2F10	10		10	AU2	610	10	DS B
AMAD2	74151N	AU2J03	10		10	AU2	903	10	DS B
AMAD2	74151N	AU2K13	10		10	AU2	1013	10	DS B
AMAD2	74151N	AU2K08	10		10	AU2	1008	10	DS B
AMAD2	74151N	AU2R11	10		10	AU2	1511	10	DS B
AMAD2	74151N	AU2M03	10		10	AU2	1203	10	DS B
AMAD2	74151N	AU2N04	10		10	AU2	1304	10	DS B
AMAD2	74151N	AU2L04	10		10	AU2	1104	10	DS B

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
AMAD2	74151N	AU2L12	10		10	AU2	1112	10	DS B
AMAD2	74151N	AU2P07	10		10	AU2	1407	10	DS B
AMAD2	74151N	AU2L05	10		10	AU2	1105	10	DS B
AMAD2	74151N	AU2M11	10		10	AU2	1211	10	DS B
AMAD2	74151N	AU2M09	10		10	AU2	1209	10	DS B
AMAD2	74151N	AU2L10	10		10	AU2	1110	10	DS B
AMAD2	74151N	AU2R13	10		10	AU2	1513	10	DS B
AMAD2	74151N	AU2R08	10		10	AU2	1508	10	DS B
AMAD2	74151N	AU2R10	10		10	AU2	1510	10	DS B
AMAD2	74151N	AU2N08	10		10	AU2	1308	10	DS B
AMAD2	74151N	AU2P09	10		10	AU2	1409	10	DS B
AMAD2	74151N	AU2P08	10		10	AU2	1408	10	DS B
AMAD2	74151N	AU2D10	10		10	AU2	410	10	DS B
AMAD2	74151N	AU2D11	10		10	AU2	411	10	DS B
AMAD2	74151N	AU2D12	10		10	AU2	412	10	DS B
AMAD2	74151N	AU2D09	10		10	AU2	409	10	DS B
AMAD2	74151N	AU2D08	10		10	AU2	408	10	DS B
					620 NET LOADS				
AMAD2A-	74S00N	AU2J19	3	*	120	AU2	919	3	AOUT
AMAD2A-	74S40N	AU2G14	9		-10	AU2	714	9	INA2
					110 NET LOADS				
AMAD2B-	74S00N	AU2G11	8	*	120	AU2	711	8	COUT
AMAD2B-	74S40N	AU2G14	10		-10	AU2	714	10	INB2
					110 NET LOADS				
AMAD2C-	74S00N	AU2H21	11	*	120	AU2	821	11	DOUT
AMAD2C-	74S40N	AU2G14	12		-10	AU2	714	12	INC2
					110 NET LOADS				
AMAD3	74S40N	AU2G14	6	*	300	AU2	714	6	OUT1
AMAD3	74151N	AU2K11	11		10	AU2	1011	11	DS A
AMAD3	74151N	AU2J12	11		10	AU2	912	11	DS A
AMAD3	74151N	AU2J03	11		10	AU2	903	11	DS A
AMAD3	74151N	AU2K08	11		10	AU2	1008	11	DS A
AMAD3	74151N	AU2K13	11		10	AU2	1013	11	DS A
AMAD3	74151N	AU2J04	11		10	AU2	904	11	DS A
AMAD3	74151N	AU2F10	11		10	AU2	610	11	DS A
AMAD3	74151N	AU2F09	11		10	AU2	609	11	DS A
AMAD3	74151N	AU2J05	11		10	AU2	905	11	DS A
AMAD3	74151N	AU2J10	11		10	AU2	910	11	DS A
AMAD3	74151N	AU2K04	11		10	AU2	1004	11	DS A
AMAD3	74151N	AU2N08	11		10	AU2	1308	11	DS A
AMAD3	74151N	AU2R10	11		10	AU2	1510	11	DS A
AMAD3	74151N	AU2M11	11		10	AU2	1211	11	DS A
AMAD3	74151N	AU2L12	11		10	AU2	1112	11	DS A
AMAD3	74151N	AU2P09	11		10	AU2	1409	11	DS A
AMAD3	74151N	AU2L04	11		10	AU2	1104	11	DS A

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
AMAD3	74151N	AU2N04	11		10	AU2	1304	11	DS A
AMAD3	74151N	AU2R13	11		10	AU2	1513	11	DS A
AMAD3	74151N	AU2P08	11		10	AU2	1408	11	DS A
AMAD3	74151N	AU2R08	11		10	AU2	1508	11	DS A
AMAD3	74151N	AU2L10	11		10	AU2	1110	11	DS A
AMAD3	74151N	AU2R11	11		10	AU2	1511	11	DS A
AMAD3	74151N	AU2P07	11		10	AU2	1407	11	DS A
AMAD3	74151N	AU2L05	11		10	AU2	1105	11	DS A
AMAD3	74151N	AU2M09	11		10	AU2	1209	11	DS A
AMAD3	74151N	AU2M03	11		10	AU2	1203	11	DS A
AMAD3	74151N	AU2D09	11		10	AU2	409	11	DS A
AMAD3	74151N	AU2D12	11		10	AU2	412	11	DS A
AMAD3	74151N	AU2D10	11		10	AU2	410	11	DS A
AMAD3	74151N	AU2D11	11		10	AU2	411	11	DS A
AMAD3	74151N	AU2D08	11		10	AU2	408	11	DS A
					620 NET LOADS				
AMAD3A-	74S40N	AU2G14	1		-10	AU2	714	1	INA1
AMAD3A-	74S00N	AU2H21	6	*	120	AU2	821	6	BOUT
					110 NET LOADS				
AMAD3B-	74S40N	AU2G14	2		-10	AU2	714	2	INB1
AMAD3B-	74S00N	AU2G11	11	*	120	AU2	711	11	DOUT
					110 NET LOADS				
AMAD3C-	74S00N	AU2H21	3	*	120	AU2	821	3	AOUT
AMAD3C-	74S40N	AU2G14	4		-10	AU2	714	4	INC1
					110 NET LOADS				
AMS0-	74151N	AU2D12	6	*	100	AU2	412	6	OUT-
AMS0-	74S00N	AU2E10	10		-10	AU2	510	10	INB3
					90 NET LOADS				
AMS1-	74151N	AU2D11	6	*	100	AU2	411	6	OUT-
AMS1-	74S0CN	AU2E10	13		-10	AU2	510	13	INB4
					90 NET LOADS				
AMS10-	74151N	AU2K04	6	*	100	AU2	1004	6	OUT-
AMS10-	74S00N	AU2H11	10		-10	AU2	811	10	INB3
					90 NET LOADS				
AMS11-	74S00N	AU2J06	5		-10	AU2	906	5	INB2
AMS11-	74151N	AU2J05	6	*	100	AU2	905	6	OUT-
					90 NET LOADS				
AMS12-	74S00N	AU2L06	2		-10	AU2	1106	2	INB1
AMS12-	74151N	AU2L05	6	*	100	AU2	1105	6	OUT-
					90 NET LOADS				

PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
AMS13-	74151N	AU2M03	6	*	100	AU2	1203	6	OUT-
AMS13-	74S00N	AU2J06	10		-10	AU2	906	10	INB3
					90 NET LOADS				
AMS14-	74151N	AU2N04	6	*	100	AU2	1304	6	OUT-
AMS14-	74S00N	AU2L06	13		-10	AU2	1106	13	INB4
					90 NET LOADS				
AMS15-	74151N	AU2L04	6	*	100	AU2	1104	6	OUT-
AMS15-	74S00N	AU2L06	10		-10	AU2	1106	10	INB3
					90 NET LOADS				
AMS2-	74S00N	AU2E10	5		-10	AU2	510	5	INB2
AMS2-	74151N	AU2D09	6	*	100	AU2	409	6	OUT-
					90 NET LOADS				
AMS3-	74S00N	AU2E10	2		-10	AU2	510	2	INB1
AMS3-	74151N	AU2D10	6	*	100	AU2	410	6	OUT-
					90 NET LOADS				
AMS4-	74S00N	AU2G11	2		-10	AU2	711	2	INB1
AMS4-	74151N	AU2D08	6	*	100	AU2	408	6	OUT-
					90 NET LOADS				
AMS5-	74S00N	AU2G11	5		-10	AU2	711	5	INB2
AMS5-	74151N	AU2F10	6	*	100	AU2	610	6	OUT-
					90 NET LOADS				
AMS6-	74S00N	AU2H11	2		-10	AU2	811	2	INB1
AMS6-	74151N	AU2F09	6	*	100	AU2	609	6	OUT-
					90 NET LOADS				
AMS7-	74S00N	AU2H11	5		-10	AU2	811	5	INB2
AMS7-	74151N	AU2J10	6	*	100	AU2	910	6	OUT-
					90 NET LOADS				
AMS8-	74S00N	AU2J06	2		-10	AU2	906	2	INB1
AMS8-	74151N	AU2J04	6	*	100	AU2	904	6	OUT-
					90 NET LOADS				
AMS9-	74151N	AU2J03	6	*	100	AU2	903	6	OUT-
AMS9-	74S00N	AU2J06	13		-10	AU2	906	13	INB4
					90 NET LOADS				
APICYCRQ	74H04N	AU2H02	2	*	120	AU2	802	2	OUT1
APICYCRQ	74H51N	AU2D03	4		-12	AU2	403	4	INC2
					108 NET LOADS				
APICYCRQ-	74H04N	AU2H02	1		-12	AU2	802	1	IN1

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
APICYCRQ-	CB	AUXPORT	64	CN	0	9999	10P1/801	64	CONN
APICYCRQ-	CB	AU2P2	62	CN	0	9999	08P2/302	62	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
APIMAINC	74H04N	AU2H02	4	*	120	AU2	802	4	OUT2
APIMAINC	74H54N	AU2G18	5		-12	AU2	718	5	INF
					108 NET LOADS				
APIMAINC-	74H04N	AU2H02	3		-12	AU2	802	3	IN2
APIMAINC-	CB	AUXPORT	39	CN	0	9999	10P1/801	39	CONN
APIMAINC-	CB	AU2P2	64	CN	0	9999	08P2/302	64	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
APIMALD	74H04N	AU2H02	6	*	120	AU2	802	6	OUT3
APIMALD	74H51N	AU2D02	13		-12	AU2	402	13	INB1
					108 NET LOADS				
APIMALD-	74H04N	AU2H02	5		-12	AU2	802	5	IN3
APIMALD-	CB	AU2P1	12	CN	0	9999	08P1/702	12	CONN
APIMALD-	CB	AUXPORT	29	CN	0	9999	10P1/801	29	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
APISTORE-	74H51N	AU2D02	4		-12	AU2	402	4	INC2
APISTORE-	CB	AUXPORT	38	CN	0	9999	10P1/801	38	CONN
APISTORE-	CB	AU2P2	41	CN	0	9999	08P2/302	41	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
APISTO-	7432N	AU2L03	1		-10	AU2	1103	1	INA1
APISTO-	CB	AU2P1	19	CN	0	9999	08P1/702	19	CONN
APISTO-	CB	AUXPORT	21	CN	0	9999	10P1/801	21	CONN
					-10 NET LOADS *EXTERNAL SOURCE*				
APIST1-	7432N	AU2N01	4		-10	AU2	1301	4	INA2
APIST1-	CB	AU2P1	53	CN	0	9999	08P1/702	53	CONN
APIST1-	CB	AUXPORT	23	CN	0	9999	10P1/801	23	CCNN
					-10 NET LOADS *EXTERNAL SOURCE*				
APIST2-	7432N	AU2N01	12		-10	AU2	1301	12	INA4
APIST2-	CB	AU2P1	51	CN	0	9999	08P1/702	51	CONN
APIST2-	CB	AUXPORT	25	CN	0	9999	10P1/801	25	CONN
					-10 NET LOADS *EXTERNAL SOURCE*				
APISOD-	7432N	AU2L03	3	*	100	AU2	1103	3	OUT1
APISOD-	74H20N	AU2H16	5		-12	AU2	816	5	IND1
					88 NET LOADS				
APISO1CK-	7432N	AU2N01	9		-10	AU2	1301	9	INA3
APISO1CK-	CB	AU2P1	57	CN	0	9999	08P1/702	57	CONN
APISO1CK-	CB	AUXPORT	75	CN	0	9999	10P1/801	75	CONN

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
					-10 NET LOADS *EXTERNAL SOURCE*				
APIS01EN-	74H10N	AU2P14	5		-12	AU2	1414	5	INC2
APIS01EN-	7432N	AU2N01	8	*	100	AU2	1301	8	OUT3
					88 NET LOADS				
APIS1D-	7432N	AU2N01	6	*	100	AU2	1301	6	OUT2
APIS1D-	7410N	AU2M12	11		-10	AU2	1212	11	INC3
					90 NET LOADS				
APIS2CK-	7432N	AU2N01	1		-10	AU2	1301	1	INA1
APIS2CK-	CB	AU2P1	49	CN	0	9999	08P1/702	49	CONN
APIS2CK-	CB	AUXPORT	72	CN	0	9999	10P1/801	72	CONN
					-10 NET LOADS *EXTERNAL SOURCE*				
APIS2D-	74H30N	AU2L15	11		-12	AU2	1115	11	ING
APIS2D-	7432N	AU2N01	11	*	100	AU2	1301	11	OUT4
					88 NET LOADS				
APIS2EN-	7432N	AU2N01	3	*	100	AU2	1301	3	OUT1
APIS2EN-	74H30N	AU2F12	11		-12	AU2	612	11	ING
					88 NET LOADS				
APIO	74H00N	AU2B02	3	*	120	AU2	202	3	OUT1
APIO	74H11N	AU2C03	5		-12	AU2	303	5	INC2
					108 NET LOADS				
APIO-	74H00N	AU2B02	2		-12	AU2	202	2	INB1
APIO-	CB	AUXPORT	6	CN	0	9999	10P1/801	6	CONN
APIO-	CB	AU2P2	9	CN	0	9999	08P2/302	9	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API1	74H00N	AU2A01	3	*	120	AU2	101	3	OUT1
API1	74H11N	AU2A02	13		-12	AU2	102	13	INC1
					108 NET LOADS				
API1-	74H00N	AU2A01	2		-12	AU2	101	2	INB1
API1-	CB	AUXPORT	7	CN	0	9999	10P1/801	7	CONN
API1-	CB	AU2P2	16	CN	0	9999	08P2/302	16	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API10	74H11N	AU2F03	5		-12	AU2	603	5	INC2
API10	74H00N	AU2E01	6	*	120	AU2	501	6	OUT2
					108 NET LOADS				
API10-	74H00N	AU2E01	5		-12	AU2	501	5	INB2
API10-	CB	AUXPORT	61	CN	0	9999	10P1/801	61	CONN
API10-	CB	AU2P2	57	CN	0	9999	08P2/302	57	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
API11	74H00N	AU2B02	6	*	120	AU2	202	6	OUT2
API11	74H11N	AU2C02	11		-12	AU2	302	11	INC3
					108 NET LOADS				
API11-	74H00N	AU2B02	5		-12	AU2	202	5	INB2
API11-	CB	AUXPORT	62	CN	0	9999	10P1/801	62	CONN
API11-	CB	AU2P2	15	CN	0	9999	08P2/302	15	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API12	74H11N	AU2F02	5		-12	AU2	602	5	INC2
API12	74H00N	AU2E01	8	*	120	AU2	501	8	OUT3
					108 NET LOADS				
API12-	74H00N	AU2E01	10		-12	AU2	501	10	INB3
API12-	CB	AUXPORT	66	CN	0	9999	10P1/801	66	CONN
API12-	CB	AU2P2	55	CN	0	9999	08P2/302	55	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API13	74H00N	AU2E01	11	*	120	AU2	501	11	OUT4
API13	74H11N	AU2F03	13		-12	AU2	603	13	INC1
					108 NET LOADS				
API13-	74H00N	AU2E01	13		-12	AU2	501	13	INB4
API13-	CB	AUXPORT	67	CN	0	9999	10P1/801	67	CONN
API13-	CB	AU2P2	53	CN	0	9999	08P2/302	53	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API14	74H11N	AU2G02	5		-12	AU2	702	5	INC2
API14	74H00N	AU2G04	6	*	120	AU2	704	6	OUT2
					108 NET LOADS				
API14-	74H00N	AU2G04	5		-12	AU2	704	5	INB2
API14-	CB	AUXPORT	69	CN	0	9999	10P1/801	69	CONN
API14-	CB	AU2P2	34	CN	0	9999	08P2/302	34	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API15	74H11N	AU2C02	5		-12	AU2	302	5	INC2
API15	74H00N	AU2A01	8	*	120	AU2	101	8	OUT3
					108 NET LOADS				
API15-	74H00N	AU2A01	10		-12	AU2	101	10	INB3
API15-	CB	AUXPORT	71	CN	0	9999	10P1/801	71	CONN
API15-	CB	AU2P2	21	CN	0	9999	08P2/302	21	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API2	74H00N	AU2A01	11	*	120	AU2	101	11	OUT4
API2	74H11N	AU2A02	11		-12	AU2	102	11	INC3
					108 NET LOADS				

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PIN SIGNATURE	CIRCUIT TYPE	DEVICE NAME	PIN NAME	PIN FUNCTION	N+ LOAD	BLOCK NUMBER	LOCATION NAME	PIN NAME	WIRING COMMENT
API2-	74H00N	AU2A01	13		-12	AU2	101	13	INB4
API2-	CB	AUXPORT	19	CN	0	9999	10P1/801	19	CONN
API2-	CB	AU2P2	18	CN	0	9999	08P2/302	18	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API3	74H11N	AU2A02	5		-12	AU2	102	5	INC2
API3	74H00N	AU2A01	6	*	120	AU2	101	6	OUT2
					108 NET LOADS				
API3-	74H00N	AU2A01	5		-12	AU2	101	5	INB2
API3-	CB	AUXPORT	41	CN	0	9999	10P1/801	41	CONN
API3-	CB	AU2P2	12	CN	0	9999	08P2/302	12	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API4	74H00N	AU2B02	11	*	120	AU2	202	11	OUT4
API4	74H11N	AU2C02	13		-12	AU2	302	13	INC1
					108 NET LOADS				
API4-	74H00N	AU2B02	13		-12	AU2	202	13	INB4
API4-	CB	AUXPORT	42	CN	0	9999	10P1/801	42	CONN
API4-	CB	AU2P2	10	CN	0	9999	08P2/302	10	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API5	74H00N	AU2E02	6	*	120	AU2	502	6	OUT2
API5	74H11N	AU2F02	13		-12	AU2	602	13	INC1
					108 NET LOADS				
API5-	74H00N	AU2E02	5		-12	AU2	502	5	INB2
API5-	CB	AUXPORT	50	CN	0	9999	10P1/801	50	CONN
API5-	CB	AU2P2	56	CN	0	9999	08P2/302	56	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API6	74H00N	AU2E02	3	*	120	AU2	502	3	OUT1
API6	74H11N	AU2E03	5		-12	AU2	503	5	INC2
					108 NET LOADS				
API6-	74H00N	AU2E02	2		-12	AU2	502	2	INB1
API6-	CB	AUXPORT	51	CN	0	9999	10P1/801	51	CONN
API6-	CB	AU2P2	52	CN	0	9999	08P2/302	52	CONN
					-12 NET LOADS *EXTERNAL SOURCE*				
API7	74H00N	AU2E02	8	*	120	AU2	502	8	OUT3
API7	74H11N	AU2F02	11		-12	AU2	602	11	INC3
					108 NET LOADS				
API7-	74H00N	AU2E02	10		-12	AU2	502	10	INB3
API7-	CB	AUXPORT	53	CN	0	9999	10P1/801	53	CONN
API7-	CB	AU2P2	58	CN	0	9999	08P2/302	58	CONN

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